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Detecting Suicide Ideation on Social Media Using DistilBERT

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Abstract

In this study, an artificial intelligence model was developed for the detection of suicidal ideation using social media data. In particular, user posts collected under the subreddits called SuicideWatch and Depression on the Reddit platform were classified using the DistilBERT model. The results obtained in the study show that the model can successfully detect suicidal ideation with a high accuracy rate (97.96%). The performance metrics of the model, such as accuracy, precision, recall and F1 score, were evaluated and it was emphasised that such models have significant potential in the field of suicide prevention but since the model is based only on Reddit's data, it has some limitations in generalisation. For future studies, it is recommended to use wider and more diverse data sources and increase the explainability of the model.

Keywords: suicide, detection, social media, transformers, distilbert, binary classification

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1. Introduction

Suicide, a global mental health issue affecting millions of individuals, is estimated to cause around 800,000 deaths annually (WHO). It stems from a complex interplay of health, environmental, and historical factors (Vioules et al., 2018). Individuals often face suicidal ideation and attempts due to an inability to cope with factors such as mental illnesses, substance abuse, and significant life stressors, necessitating urgent mental health support (Bhaumik et al., 2023). Generally, individuals fear societal stigma and choose to conceal their suicidal thoughts and attempts, posing a significant challenge in suicide detection and prevention. However, research suggests that the intensity of individuals' fear of social stigmatization and exclusion tends to be lower in online communities (Heckler, de Carvalho and Barbosa, 2022). Consequently, individuals prone to suicide and harbouring suicidal thoughts are increasingly expressing themselves openly on various social media platforms. To provide timely support to individuals struggling with suicidal thoughts, professionals are developing various artificial intelligence models. These models can identify suicide risk-laden posts and alert relevant authorities or individuals for immediate assistance.

AI models can play a crucial role in early detection, risk assessment, and effective interventions. By analyzing vast datasets from diverse social media platforms, they can predict suicidal ideation, even providing personalized support by tracking individuals' online behaviour (Tighe et al., 2017) or identifying cues indicative of suicidal thoughts through emotional content analysis (Sawhney, Joshi and Flek, 2021). In recent years, the accuracy rate of artificial intelligence models built to predict suicidal behaviour has been higher than traditional methods (Heckler, de Carvalho and Barbosa, 2022). For example, in a study conducted with a dataset consisting of university students, it was seen that it predicted suicidal thoughts and behaviour using the Random Forest model, and the best prediction score was obtained (AUC = 0.8) with cross-validation in this model, which included suicidal thoughts, trait anxiety, depression symptoms, and self-esteem

variables (Macalli et al., 2021). While deep learning techniques excel at identifying complex patterns and relationships, these models often operate as black boxes, requiring improvement in explainability and interpretability. This is because the classification decisions made by deep learning models need to be highly transparent. To achieve this, more diverse datasets and bias mitigation techniques are essential for model development and this proposal intends to explore the latest technologies to achieve a better prediction of suicide ideation with data derived from social media platforms.

2. Background

In the psychology literature, the factors contributing to suicidal ideation include perceived burdensomeness and thwarted belongingness, defeat and entrapment, and the combination of pain and hopelessness. Research indicates that Dialectical Behavioural Therapy (DBT) is effective in individuals with suicidal thoughts (Klonsky, May and Saffer, 2016). However, without the disclosure of suicidal thoughts and past attempts, it is challenging to detect those individuals to provide appropriate help. Although professionals have recommended the restriction of means used in suicides, school-based psychoeducation programmes, and social support systems to prevent suicidal thoughts and attempts, predicting those individuals at the earliest possible phase is not feasible if the individual is not examined in a clinical setting (Klonsky, May and Saffer, 2016). Indeed, a study revealed that 60% of men who died by suicide had no prior record of mental illness, underscoring that suicide detection should not be limited to individuals with a history of mental illness and that identifying suicidal ideation is a significantly challenging task (Ji, 2022).

Therefore professionals have opted for an interdisciplinary approach, leveraging artificial intelligence technologies for this purpose. Currently, machine learning models and deep learning models are being developed to detect suicide ideation by analysing data from social media, electronic health records, and suicide notes. In the realm of machine learning models, Support Vector Machines, Logistic Regression, Random Forests, Naive

Bayes, and XGBoost have been implemented using engineered features such as word frequencies, sentiment scores, and linguistic markers (D'Hotman and Loh, 2020). On the deep learning side, models like Convolutional Neural Networks (CNN) and hybrid CNN-LSTM models have been utilised, and research indicates the superior performance of deep learning models compared to machine learning models (Aldhyani et al., 2022). Another study demonstrated that CNN-based deep learning models developed with electronic health records data exhibit higher predictive performance than traditional machine learning models (Pigoni et al., 2024).

However, several challenges need to be addressed to advance the development of deep learning models for detecting suicide ideation. According to the current literature, key issues include the lack of large, diverse, and well-annotated data, lack of interpretability and explainability in deep learning models, the failure to capture context and intent, leading to potential false positives or negatives, and ethical and privacy concerns arising from the use of data from social media or electronic health records (Ji et al., 2021) (Marks, 2019).

One study proposed incorporating clinical questionnaires to help models capture context and intent better, thereby enhancing the ability of NLP models to understand the underlying intent of social media posts (Ji, 2022). In 2023, a performance comparison was conducted among Transformer-Based Language Models (TLMs) such as mentalBERT, mental-RoBERTa, BERT, RoBERTa, BioBERT, and ClinicalBERT. The pre-trained TLMs on mental health data showed high performance in F1 and recall scores (Greco et al., 2023). Similarly, the PsychBERT model, pre-trained on mental health texts, demonstrated high performance in binary and multi-class classification when trained with social media data, highlighting the need for multi-modal or cross-modal analysis using TLM models with social media and EHR data (Greco et al., 2023).

To enhance the explainability of deep learning models, research is conducted to create a text-to-text pipeline that allows the calculation of

overlap statistics concerning the ground truth, and suggests that these explanations should be validated by health professionals (Souto, Pérez and Parapar, 2023). Another study recommended the use of sentiment analysis and pre-trained language models to overcome the lack of intention understanding which is a common issue with deep learning models (Ji, 2022). In this context, the previously mentioned PsychBERT model, trained on diverse mental health datasets, has been introduced into the literature, and the utilisation of social media data has been proposed to improve intention understanding due to the inclusion of various intentions and emotional states within large datasets (Vajre et al., 2021).

3. Aims and Objectives

This project aims to develop a highly accurate, interpretable and domain-adapted DistilBERT model to improve early detection of suicide ideation in labelled unstructured data from a social media platform called Reddit.

- Conduct a thorough literature review and evaluate different deep learning and machine learning models implemented in various techniques and approaches for suicide ideation detection.
- Explore the potential of social media data to excel in a suicide ideation classification task.
- To adapt DistilBERT's language knowledge to the data distribution of suicide ideation text, fine-tune the transformers model and evaluate the performance metrics.

4. Literature Review

Over the years, professionals have developed many machine learning and deep learning models with various techniques to analyse data such as electronic health records, suicide notes, clinical interviews, and social media content. For instance, mobile technologies such as iBobbly were developed to assess the effectiveness of mobile applications in reducing suicidal ideation among Indigenous people in Australia and although the experiment was conducted with a small sample size ($N = 61$), results have shown a significant decrease in depression and psychological distress (Tighe et al., 2017).

Also, IBM developed several technologies such as automated speech analysis to monitor the mental states of patients, and a study found that the use of Woebot (cognitive-behavioural therapy-based chatbot) significantly reduced depression symptoms in users (Kalandarian and Nasrallah, 2019) and sensing technological devices used by people can be utilised for suicide ideation detection to detect the most common depression symptoms that cause suicidal ideation. In this regard, a mobile tool called Boamente uses a virtual keyboard application to analyse the texts written by patients on their keyboards with artificial intelligence and It prepares graphs about those patients' suicidal thoughts for professionals to act on (De Oliveira et al., 2022).

In another study, when neural network architecture and word embedding techniques were combined, it was observed that it gave good results in suicide ideation detection with social media data, and it was suggested to study real-time suicidal ideation detection on social media data with different deep learning models (Tadesse et al., 2019). Hence, it is clear that mobile applications are feasible methods for providing mental health support. However, mobile applications only could be useful if individuals are willing to get help which is less likely to happen when it comes to suicidal thoughts because research has shown that people with suicidal thoughts could have feelings of embarrassment and shame. Also, they could be in a denial phase

where they just want to remain positive and suppress their inner turmoil (Lebow, 2021).

In order to offer help to those who prefer not to disclose themselves, professionals need to dig deeper to address the common keywords. For instance, Huang, Goh and Liew (2007) collected and manually annotated social media data to identify bloggers with suicidal tendencies on MySapce.com. They assigned a weighting to its significance for common words found in actual suicide notes and suicide prevention websites and found that 25% of the top-ranked bloggers have shown signs of depression or suicidal tendencies.

In another study, the posts in the last 24 hours of a Twitter user who committed suicide were analysed to identify suicidal ideation patterns using linguistic inquiry and word count tools (F. Gunn and Lester, 2012). The study has shown that there is a decrease in self-reference which may reflect a disconnection from the self and life as the individual gets closer to death and it is emphasised that real-time social media analysis could provide crucial insights. In another research, it was also seen that suicide-related conversations generally included social and personal problems by conducting topic modelling and cloud visualisation (Ji et al., 2018).

Another analysis showed that online linguistic language patterns were related to the language used by depressed individuals (Rude, Gortner and Pennebaker, 2004), and it was emphasised how important these patterns were in the transition from depressive thoughts to suicidal thoughts (De Choudhury et al., 2016). Regarding this, researchers found that as individuals with suicidal thoughts approached the date of their suicide attempt, their depressive posts on Twitter increased and anger was added to this as they got even closer to attempt harming themselves (Qntfy et al., 2016). At this point, researchers developed a transformer-based model called STATENet by analysing the emotional and time relationships in Twitter users' past posts and emphasised the need for larger data sets with more linguistic and cultural diversity in studies (Sawhney et al., 2020).

Elsayed, Elsayed and Ozer (2024) have worked on a deep gate unit-based model (GRU) due to its simple architecture that makes training process faster to detect suicide ideation in Reddit posts and they found out that GRU achieved the highest performance (accuracy=94.33%) comparing to LSTM (accuracy=93.5%), CLSTM (accuracy=93.2%), BiGRU (accuracy=93.4%), BiLSTM (accuracy=93.6%) and SVC (accuracy=91.9%).

Researchers stated that transformer-based models such as BERT perform better in detecting suicidal thoughts instead of traditional and old neural networks-based models, and they emphasised that social media data is a very valuable resource in suicidal ideation detection research (Bhardwaj et al., 2022). So, in a Twitter study, researchers were able to provide a risk assessment up to the last 10 days before a suicide attempt and their model reached a 0.88 AUC score by using neural networks to analyse the connection between individuals' tweets and psychological phenomena such as loneliness, burden and stress. Although researchers have been utilising the data from social media platforms such as Twitter and Reddit, these datasets are mostly unbalanced and poor quality (Liu, Shi and Jiang, 2022). Therefore, researchers used large language models such as ChatGPT, Flan-T5, and Llama to generate synthetic data and obtained an F1 score of 0.82 with the BERT model trained with synthetic data and textual data, physiological signs or acoustic features from speech were combined for more comprehensive risk analysis to apply multimodal approaches for improving intention understanding (Ghanadian, Nejadgholi and Osman, 2024). However, research shows that using synthetic data could lack realism and accuracy and may not perfectly reflect real-world conditions (Mustać, 2024).

Apart from the problem of lack of data diversity, researchers have conducted various studies on the interpretability, explainability and intention understanding of the models. Interpretability allows us to understand how the model works and the reason behind its decisions. For instance, linear regression or decision trees are considered interpretable because those are

transparent enough to understand the decision-making processes. However, as AI models get more complex like deep neural networks, interpretability gets harder due to its intricate structure. For complex models, explainability offers a better solution by creating explanations for the decisions instead of trying to understand the model (Giosmin, 2023).

Researchers have developed several methods to address these challenges. A recent favourite approach is the analysis of attention mechanisms, which can give a clue to what the model is focusing on while making decisions. However, in contrast to traditional deep learning models, where it is still feasible to have some form of interpretation of decision processes based on feature maps or specific neuron activations, Transformer models need a much more sophisticated unpacking process to shed light on how they make decisions. For example, BERT (Bidirectional Encoder Representations from Transformers), present clear benefits of this architecture as well as its challenges (Maruthi et al., 2022).

The fact that BERT was able to process language in both directions, as opposed to left-to-right reading, helped in making it quite successful in carrying out NLP tasks. However, this bi-directionality, together with the depth of the model, makes the internal mechanisms of BERT very hard to understand. More advanced techniques, such as Layer-wise Relevance Propagation, have been used towards better interpreting how the BERT model arrives at specific predictions. Other research studies have probed into the mechanistic interpretability of the BERT models in an attempt to reverse-engineer its computations for clarity on which parts make specific decisions. Although such approaches do lead to insightful observations, they are rather complex and require more research to make them applicable on a wider scale (Rai et al., 2024).

5. Research Methodology

5.1. Mixed Methods Approach Using Machine Learning, Deep Learning and Clinical Psychology

1. Data Collection and Preparation:

- Pre-process the data derived from Kaggle (the dataset of subreddits of “Suicide Watch” and “depression” from the Reddit platform), such as noise removal, tokenization, emojis and abbreviations.

2. Development of the Model:

- Adaptation or fine-tuning of the pre-trained model DistilBERT model with the annotated Reddit dataset.

3. Feature Engineering and Selection:

- Apply techniques from the domain of natural language processing to capture language patterns indicative of suicidal thoughts.

4. Model Performance Evaluation:

- Evaluate the model using standard machine learning evaluation metrics like F1 score, accuracy, recall, and precision and apply a confusion matrix.

5. Ethical Implications:

- Considering ethics and privacy in handling sensitive mental health data on social media throughout the project.

This methodology will incorporate state-of-the-art machine learning and deep learning with insights from clinical psychology. The mixed methods approach allows for a quantitative assessment of model performance and a qualitative evaluation of its clinical relevance and interpretability.

6. Dataset Description and Preparation

To train the DistilBERT model for predicting suicide ideation, I utilised a dataset obtained from the Kaggle website, which consists of users' posts on Reddit under two subreddits called "SuicideWatch" and "depression", annotated as either "suicide" or "non-suicide". This dataset includes 232074 rows in total and two columns as "text" and "class":

- "text": This column contains raw text data, which encompasses a variety of posts and messages from individuals. Each entry is a brief narrative that may convey personal thoughts, emotions, or statements. The texts differ in length and complexity, reflecting both explicit and implicit mentions of mental health challenges.
- "class": This target label column is marked as either "suicide" or "non-suicide," indicating whether the associated text suggests suicidal thoughts. Entries labelled as "suicide" indicate a potential risk based on the language used, while those marked as "non-suicide" do not imply such risks.

The dataset is equally distributed (50/50) between the two labels, which is essential for training a model capable of accurately discerning between them. To prevent any issues related to data imbalance, It is ensured that the dataset adequately represents both categories before starting model training.

6.1. Word Clouds

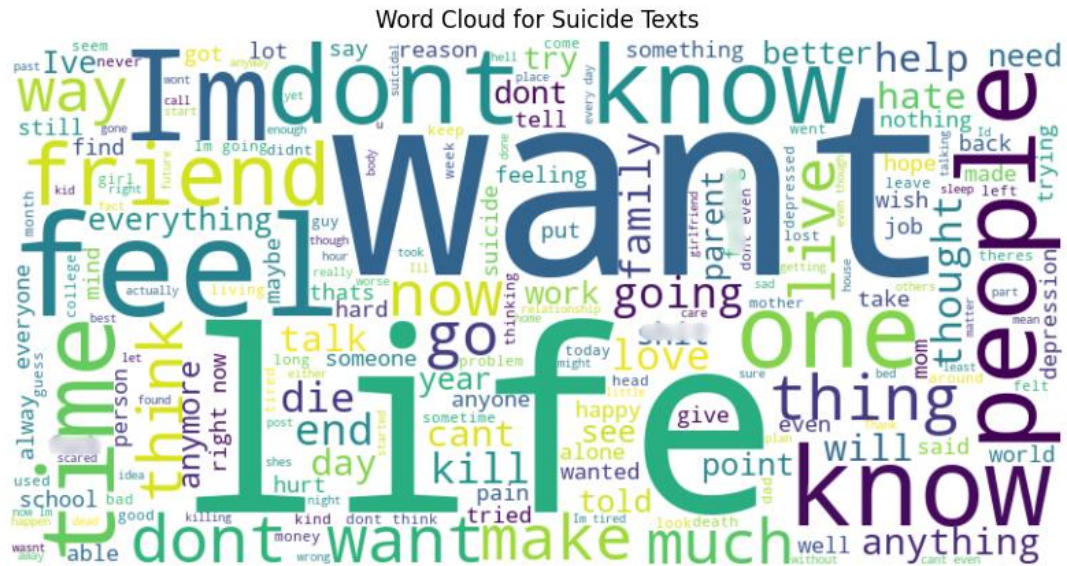


Figure 1: Word Cloud for “suicide” labelled texts in train dataset.

The word cloud for suicide-labelled texts reveals the intense emotional struggles people express. Words like "life," "want," "feel," and "know" stand out, highlighting the overwhelming focus on existential and emotional concerns. Terms like "friend," "family," and "help" show the desire for connection and support, while words like "kill" and "die" tragically indicate the mental distress people are experiencing.

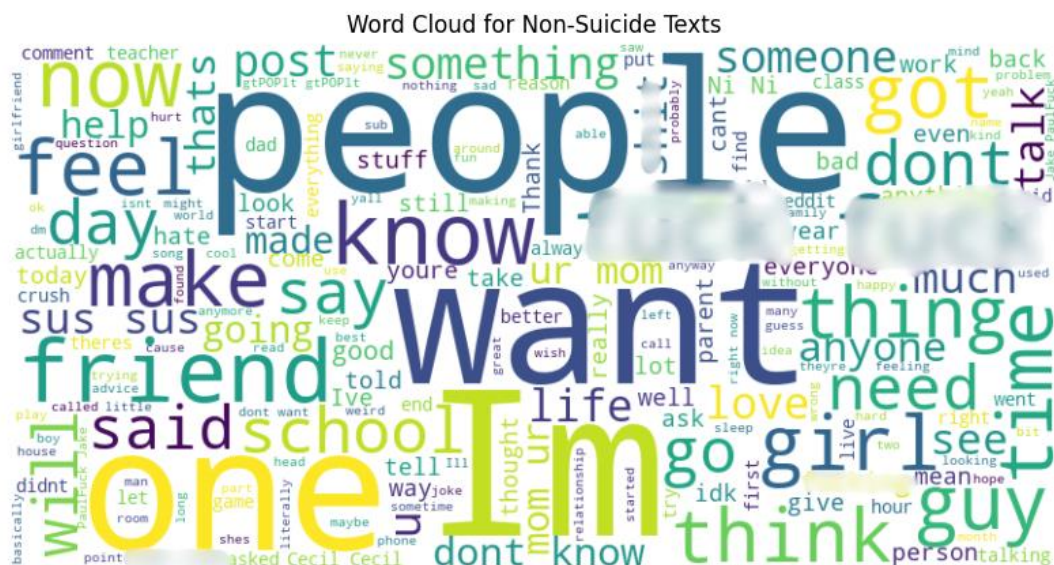


Figure 2: Word Cloud for “non-suicide” labelled texts in train dataset.

In contrast, the word cloud for non-suicide texts showcases more casual and everyday language. Words like "people," "friend," "school," and "day" emphasise the lighter, more social nature of these conversations. Although some strong language like "f***" appears frequently, it reflects informal online exchanges rather than emotional distress. Overall, the focus is on social interactions and daily life, with less emotional intensity compared to the suicide-labelled texts.

Before inputting the data into the DistilBERT model, several preprocessing steps were undertaken:

1. Text Cleaning:

- Special characters, HTML tags, and unnecessary punctuation were removed.
- Common stopwords were preserved, as they often provide contextual significance in recognising expressions of distress or ideation.

2. Tokenization:

Given that the DistilBERT model functions on tokenized text, the raw text was tokenized using the pre-trained DistilBERT tokenizer. This process segments the text into sub-word tokens, enabling the model to more effectively manage out-of-vocabulary terms and contextual subtleties.

3. Dataset Splitting:

The dataset was divided into training, validation, and test sets following a 60-20-20 split. This arrangement ensured the model had sample data for learning while maintaining a sufficient number of samples for unbiased evaluation.

This thorough dataset and preparation process established a robust foundation for accurately training the DistilBERT model to predict suicide ideation, as will be elaborated in the following sections.

7. Model Training, Testing and Evaluation

For this study, the DistilBERT model was employed to classify textual data as either "suicide" or "non-suicide." This model was developed as a lighter and faster version of the BERT model. It aims to provide similar performance using less computational resources while preserving the high achievements of BERT. DistilBERT is achieved by transferring the information learned from large language models to a smaller and more efficient model. It is transformer-based and takes into account bidirectional contexts for language models. Since it has fewer parameters compared to BERT, it provides advantages in speed and memory usage in training and inference processes. In addition, it learns the predictions of a large model using the knowledge distillation technique and thus achieves similar performance (Babulal and Nayak, 2022). This section outlines the training process, testing setup, and evaluation metrics used to assess the model's performance.

7.1. Model Training

1. Pre-trained Model:

The model was initialised with the pre-trained DistilBERT weights, which are fine-tuned for downstream tasks such as text classification. The advantage of using a pre-trained model is that it already has a general understanding of language structure, which significantly reduces the training time required to adapt the model to the specific task of suicide ideation prediction.

2. Fine-tuning:

Fine-tuning was performed by adding a classification head (a dense layer followed by a softmax activation) on top of DistilBERT, tailored to the binary classification task. During fine-tuning, the entire model, including the pre-trained layers, was trained on the suicide ideation

dataset. This allows the model to adapt its language representations to better identify nuances specific to the task.

3. Training Procedure:

- Optimiser: The AdamW optimiser was used for training. AdamW differs from the traditional Adam optimiser by decoupling the weight decay term from the gradient-based optimisation. Weight decay is a regularisation technique that helps prevent overfitting by penalising large weights. In this case, a weight decay value of $5e-4$ was used, which helps to ensure that the model's weights remain small and stable during training.
- Loss Function: Cross-entropy loss was employed, as this is a standard choice for binary classification tasks.
- Batch Size: A batch size of 8 was chosen to balance between computation time and memory constraints, given the length of the input sequences.
- Epochs: The model was trained for 5 epochs, with early stopping applied to prevent overfitting. The early stopping criterion monitored validation loss stops training if the performance on the validation set is no longer improving for 2 consecutive epochs.

7.2. Testing Setup

After the model was trained, it was evaluated on the test set, which consisted of 20% of the original dataset. The test set was kept completely unseen during the training process to ensure an unbiased evaluation of the model's performance.

Before evaluating the test data, the following pre-processing steps were applied, identical to those used during training:

- Tokenization of text using the DistilBERT tokenizer.
- Padding and truncation to ensure uniform input length.

The model was then used to predict the class of each sample in the test set, providing probability scores for each class, which were converted into binary predictions.

7.3. Evaluation Metrics

The performance of the model was evaluated using several metrics to ensure a comprehensive understanding of its strengths and weaknesses in predicting suicide ideation. These metrics include:

1. Accuracy:

Accuracy is the ratio of correctly predicted instances (both "suicide" and "non-suicide") to the total number of instances in the test set. While accuracy provides a broad measure of performance, it may not fully capture the nuances of this task, especially if the data distribution is slightly imbalanced (Abbasian et al., 2024).

2. Precision:

Precision measures the proportion of true positive predictions (correctly identified suicide ideation instances) out of all the instances predicted as positive. It is a crucial metric for this task, as high precision ensures that the model is not over-triggering false alarms when identifying suicide ideation (Abbasian et al., 2024).

3. Recall:

Recall is the proportion of true positive instances correctly identified by the model. In the context of suicide ideation detection, high recall ensures that the model captures as many true cases of suicide ideation as possible, minimising false negatives (Abbasian et al., 2024).

4. F1 Score:

The F1 score is the harmonic mean of precision and recall, providing a single metric that balances both false positives and false negatives. This is particularly useful for evaluating the model in cases where one class is more critical to identify than the other (Abbasian et al., 2024).

5. Confusion Matrix:

A confusion matrix was used to provide a detailed breakdown of the model's performance, showing the number of true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN). This helps to identify any patterns in the model's misclassifications, informing possible improvements (Abbasian et al., 2024).

8. Results

The performance of the model was evaluated based on several key metrics, including loss, accuracy, and the confusion matrix, to gain a comprehensive understanding of its strengths and weaknesses.

8.1. Loss and Validation Accuracy

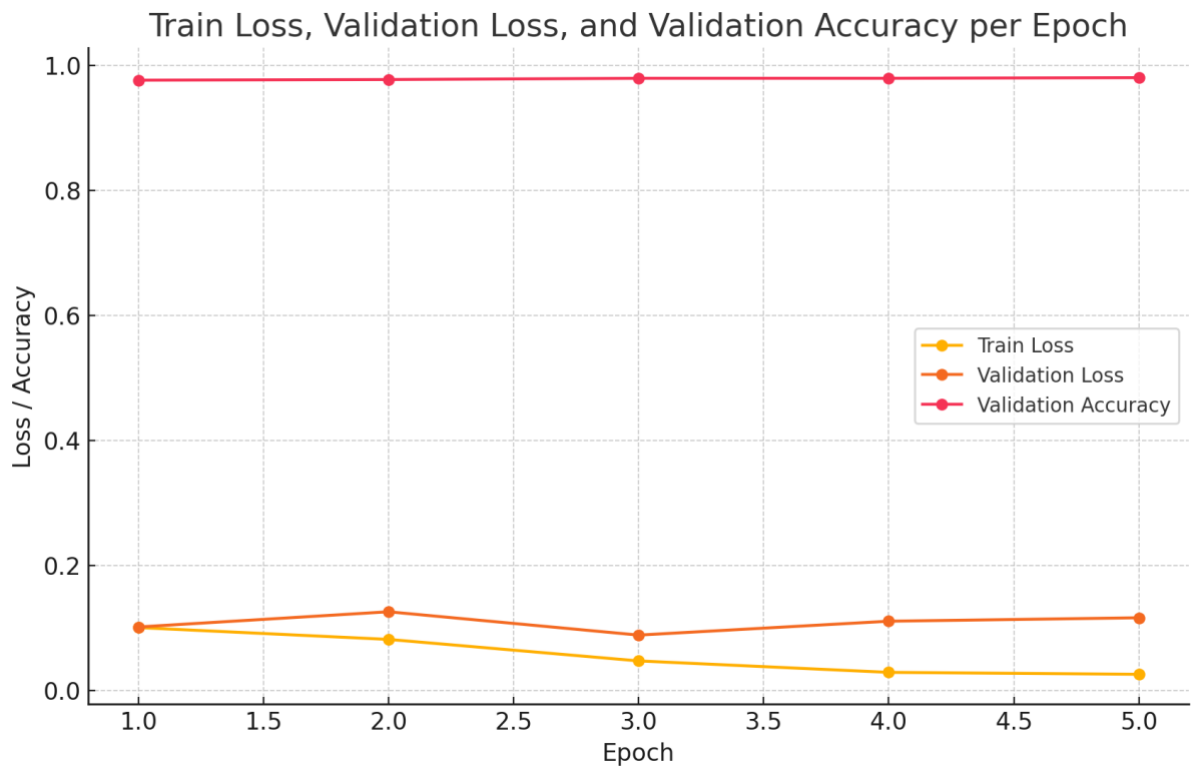


Figure 3: Graph for Train Loss, Validation loss and Validation Accuracy

The training and validation loss trends, as shown in the previous figure, indicate that the model performed well up to the third epoch, where both the training and validation loss were minimised. However, after epoch 3, the validation loss began to rise, indicating potential overfitting. However, the validation accuracy remained around 0.97 – 0.98 throughout all epochs, which suggests that the model made correct predictions for the validation set.

Given the increasing validation loss after Epoch 3, the model's version from Epoch 3 was selected for further evaluation and testing, as it

represented the optimal trade-off between minimising loss and avoiding overfitting.

8.2. Confusion Matrix

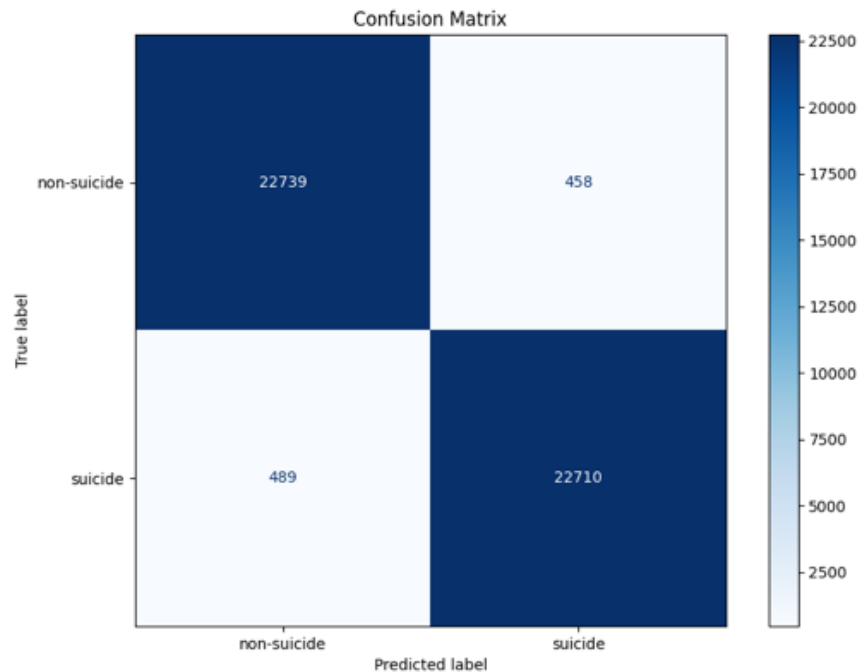


Figure 4: Confusion matrix for the DistilBERT model

The confusion matrix above shows the model’s performance on the test set, where the true labels are compared with the model’s predictions. The matrix provides the following insights:

- True Positives (TP): The model correctly identified 22,710 instances of “suicide.”
- True Negatives (TN): The model correctly identified 22,739 “non-suicide” cases.
- False Positives (FP): The model misclassified 458 “non-suicide” instances as “suicide.”
- False Negatives (FN): The model failed to detect 489 cases of “suicide,” labelling them as “non-suicide.”

These results suggest that the model is performing well in both detecting suicide ideation and identifying non-suicidal posts, but with a slight tendency toward false negatives, where some “suicide” cases are missed.

8.3. Evaluation Metrics

Based on the model’s predictions and the confusion matrix, the following evaluation metrics were calculated:

1. Accuracy:

The model achieved a high accuracy of 97.96%, indicating that it correctly classified most instances in the dataset.

2. Precision (for suicide class):

A precision of 98.02% for the suicide class shows that the model is good at minimising false positives, meaning when it predicts “suicide,” it is likely to be correct.

3. Recall (for suicide class):

The recall of 97.89% means that the model successfully captures most of the “suicide” instances, but it misses a small number of actual cases.

4. F1 Score:

The F1 score of 97.96% reflects the balance between precision and recall, suggesting that the model has a strong performance overall in detecting suicide ideation.

9. Discussion

In this study, the DistilBERT transformers model was developed to detect suicidal tendencies among social media users (Reddit) and the performance of the model was analysed with various evaluation metrics. The high accuracy achieved by the model shows that individuals with suicidal tendencies can be detected through social media data. These results reveal how effective artificial intelligence can be in such data.

Although the model showed a very high success rate with 97.96% accuracy and F1 score, the fact that the model makes a low number of false negative predictions shows that some risky individuals may be missed and that failing to identify even one high-risk individual could result in severe consequences. This situation shows that training with more data is required or that the model parameters must be improved to obtain better results.

In a similar study, it was reported that the DistilBERT model showed superior performance compared to RNN-based models, especially in suicide detection. In this study, DistilBERT achieved better results than RNN-based models such as Bi-LSTM with 97.7% accuracy (Akintoye, Wei and Liu, 2024). The results of both studies show that DistilBERT can provide fast and effective results, especially on social media data, despite its lightweight structure.

In another study, researchers Kanojia and Babulal (2022) achieved 92% accuracy with the DistilBERT model in just one epoch of training. Although this success achieved in just one epoch is impressive, it is emphasized that training the model for a longer time and adjusting the hyperparameters can further improve the model's performance. In another study, (Sharma and Karwasra, 2023) tried to detect suicidal content in social media posts using both LSTM and DistilBERT models. In their results, it was seen that the DistilBERT model outperformed LSTM and achieved 97.76% accuracy. LSTM achieved 93.26% accuracy. This result reinforces the fact that

Transformer-based models provide better performance than other architectures.

Other than the hardware limitations that is encountered for this project, one of the main limitations of this study is that the dataset used is taken only from one social media platform. This may limit the generalization capacity of the model and it may be difficult to fully evaluate the performance of the model in a broader context encountered in real life. In addition, the dataset used in this study covers certain language or cultural groups so this limitation may restrain the performance of the model in different social environments. In the future, it is thought that models trained with a much larger and possibly multimodal data pool may overcome such limitations.

Overall, this study emphasizes the potential of transformer-based models like DistilBERT in suicide prevention efforts. Future work that addresses current limitations, such as data diversity and false negative rates, will improve the real-world applicability of these models.

10. Ethical Considerations

In ethical evaluations regarding the use of artificial intelligence algorithms, the accuracy of the algorithms has been carefully considered during model development processes because false positives/negatives in classification decisions could cause significant distress to individuals. Another ethical concern is bias encountered because the dataset does not represent a variety of people from different religions, races, nationalities, etc. Additionally, artificial intelligence models should not replace human judgement and there should always be a clinical assessment for the decision-making process of those models.

11. Conclusion

In this paper, suicide ideation has been successfully detected in Reddit posts using the DistilBERT pre-trained transformer model. The experimental results show that the model achieves a good performance with 98.85%, 97.93%, and 97.96% accuracy for training, validation, and testing, respectively. These results highlight the effectiveness of transformer-based models like DistilBERT in understanding and identifying suicidal tendencies through social media data. For further research, expanding the data sources beyond Reddit would provide a better generalisation ability for the classifier and greatly benefit individuals in obtaining appropriate assistance to address suicidal ideation.

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Click the link below to access the code and the dataset used in this project:

https://github.com/mehmetfurkanakpinar/SID_DistilBERT.git